

Architectural Justification

We chose Layered Architecture for the Digital Campus Library Reservation Gateway System.

Architectural Choice

The system is organized into a Presentation Layer (Web UI Component), a Business Layer (Reservation Manager, Catalog & Loan Service, and Notification Components), and a Data/Integration Layer (Database Component and Campus SSO Adapter Component).

Reasons for Selection

1. Clear separation of concerns: Reservation logic (FIFO queueing, hold limits), catalog/loan logic (search, checkout, fines), and notification dispatch are distinct, well-understood business rules that map naturally onto separate business-layer components, while the UI and data access stay isolated from that logic — matching the structured, policy-heavy nature of the library domain (FR-001–FR-005).
2. Simplicity and maintainability for a moderate-scale system: with a single institution's catalog (10,000+ records) and up to 500 concurrent users (NFR-001), a layered monolith is easier to build, test, and deploy than a distributed microservices architecture, avoiding unnecessary network latency and operational overhead.

Security Advantage

All reservation, checkout, and account operations must pass through the Business Layer, which enforces authentication via the Campus SSO Adapter Component before reaching the Data Layer (NFR-002). This centralizes authorization checks in one layer rather than duplicating them across UI and data-access code, reducing the risk of an unauthenticated request reaching the Database Component directly.

Performance Benefit

Because the Business Layer components communicate with the Database Component through a single, well-defined DB Interface, query patterns (e.g., catalog search, availability checks) can be optimized and indexed in one place, helping meet the 2-second response time target under 500 concurrent users (NFR-001) without the added round-trip latency that a distributed service-to-service architecture would introduce.